

A Complemented-Summand Transfer for Small Gradient Cones

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Status

This packet records a candidate partial result, likely valid. It gives a permanence theorem for the gradient-cone property in Open Problem 4.5 of Azagra–Jimenez-Sevilla, arXiv:math/0002108. It does not solve the problem for all separable reflexive spaces.

Source Context

For a smooth bump b on a Banach space X , Azagra and Jimenez-Sevilla consider the cone

$$\mathcal{C}(b) = \{\lambda b'(x) : \lambda \geq 0, x \in X\} \subset X^*.$$

They note that this cone is always norm dense, and ask how small it can be. After proving that in $X = \ell_2$ there is a C^1 Lipschitz bump whose gradient cone has empty interior, they formulate the following problem.

Let $\theta : \mathbb{R} \rightarrow \mathbb{R}$ be a C^1 bump function with empty interior of its gradient cone. Then θ is not Lipschitz on bounded sets. $\square$

It is clear that the proof of the preceding theorem could only be adapted for superreflexive Banach spaces X which admit a decomposition as an infinite sum of subspaces isomorphic to X , and therefore the problem whether in a separable reflexive space there can be bumps whose sets of gradients have empty interior remains open in the general case, though the information that has already been gathered seems to point to a final positive solution.

Open Problem 4.5. Let X be a (separable reflexive) infinite-dimensional Banach space which admits a C^p smooth (Lipschitz) bump function. Is there another C^p smooth (Lipschitz) bump function f on X so that the cone $\mathcal{C}(f) = \{\lambda f'(x) : \lambda \geq 0, x \in X\}$ has empty interior in X^* ?

Result

Theorem 1 (Complemented-summand transfer). *Let $X = Y \oplus Z$ be a Banach-space direct sum. Suppose that Y admits a C^p smooth Lipschitz bump $b : Y \rightarrow \mathbb{R}$ such that $\mathcal{C}(b)$ has empty interior in Y^* , and that Z admits a nonnegative C^p smooth Lipschitz bump $\eta : Z \rightarrow \mathbb{R}$. Then X admits a C^p smooth Lipschitz bump $F : X \rightarrow \mathbb{R}$ such that $\mathcal{C}(F)$ has empty interior in X^* .*

Proof. Choose $y_0 \in Y$ and $z_0 \in Z$ with $b(y_0) \neq 0$ and $\eta(z_0) > 0$. Define

$$F(y, z) = b(y)\eta(z), \quad (y, z) \in Y \oplus Z.$$

This is a nonzero C^p smooth Lipschitz bump: its support is contained in $\text{supp } b \times \text{supp } \eta$, hence is bounded in the direct-sum norm, and products of bounded Lipschitz functions with bounded support are Lipschitz.

Under the canonical identification $X^* = Y^* \oplus Z^*$, the derivative is

$$F'(y, z) = (\eta(z)b'(y), b(y)\eta'(z)).$$

Let $P : X^* \rightarrow Y^*$ be the continuous open projection. Since $\eta \geq 0$, for every $\lambda \geq 0$ one has

$$P(\lambda F'(y, z)) = \lambda \eta(z)b'(y) \in \mathcal{C}(b).$$

Thus $P(\mathcal{C}(F)) \subseteq \mathcal{C}(b)$. If $\mathcal{C}(F)$ had nonempty interior in X^* , then, because P is an open surjection, $P(\mathcal{C}(F))$ would have nonempty interior in Y^* . This contradicts the assumption that $\mathcal{C}(b)$ has empty interior. Therefore $\mathcal{C}(F)$ has empty interior in X^* . $\square$

Corollary 1 (Complemented Hilbert subspaces). *Let X be a Banach space with a C^1 smooth Lipschitz bump. If X contains a complemented subspace isomorphic to ℓ_2 , then X satisfies the C^1 Lipschitz-bump conclusion of Open Problem 4.5: it has a C^1 smooth Lipschitz bump whose derivative cone has empty interior in X^* .*

Proof. The source paper proves the required small-cone bump on ℓ_2 . Transport that bump through the linear isomorphism from the complemented subspace Y onto ℓ_2 ; empty interior is preserved by the adjoint isomorphism.

Write $X = Y \oplus Z$. It remains only to get a bump on Z . Let $\beta : X \rightarrow \mathbb{R}$ be a nonzero C^p smooth Lipschitz bump on X , and choose a point $x_0 = y_0 + z_0$ with $\beta(x_0) \neq 0$. Then

$$\eta(z) = \beta(y_0, z + z_0)^2, \quad z \in Z,$$

is a nonnegative nonzero C^p smooth Lipschitz bump on Z . Apply Theorem 1. $\square$

Proof Intuition

The construction uses a product bump. The derivative in the solved summand Y is only multiplied by a nonnegative scalar coming from the other factor, so the Y^* -projection of the new derivative cone stays inside the old small cone $\mathcal{C}(b)$. Since projections from $Y^* \oplus Z^*$ onto Y^* are open maps, a cone with interior upstairs would project to a set with interior downstairs. This is impossible.

Implication for Open Problem 4.5

Open Problem 4.5 remains open for arbitrary separable reflexive spaces. The transfer theorem shows that the source's Hilbert-space construction is not isolated: it automatically solves the problem for every smooth space with a complemented Hilbert summand, and more generally for every complemented extension of a space already known to have a small derivative cone bump.

Novelty and Literature Check

The local run indexes were searched for 0002108, Rolle theorem phrases, gradient-cone phrases, complemented summands, and empty-interior gradient-cone terms. No duplicate packet or attempt was found. Bounded web search for the exact phrases “sets of gradients have empty interior” and

“cone generated by their derivatives has empty interior” found the source paper but no later full solution. Local later-source search found arXiv:2407.21036 on lineability of Rolle-failing bumps, but not an answer to Open Problem 4.5.

The result is elementary enough that it may be folklore; the packet records it because it is a useful reduction for future attempts and gives a clean new family beyond the literal Hilbert-space statement in the source paper.

References

- [1] D. Azagra and M. Jimenez-Sevilla, *Rolle’s theorem is either false or trivial in infinite-dimensional Banach spaces*, arXiv:math/0002108, 2000.